

Wholesale Customer Segmentation

Project 03 — Unsupervised Learning | MenuManager ML Internship

Intern	Darshan Suresh S
Dataset	UCI Wholesale Customers (440 records)
Tools	Pandas, NumPy, Matplotlib, Scikit-learn, SciPy
Method	K-Means (k=3), validated against Agglomerative Clustering

1. Problem Statement

A wholesale distributor serves customers with very different purchasing patterns. Without a labelled target, this project builds a data-driven unsupervised workflow that discovers meaningful customer segments from annual spending across six product categories (Fresh, Milk, Grocery, Frozen, Detergents_Paper, Delicassen) for 440 customers, and translates the clusters into actionable business profiles.

2. Method

2.1 Exploratory Analysis

All six spending categories showed severe right-skew (skewness ranging 2.55 to 11.11), with Delicassen the most extreme. In every column the mean exceeds the median and the maximum is far above the 75th percentile, indicating a small number of very large-spending customers.

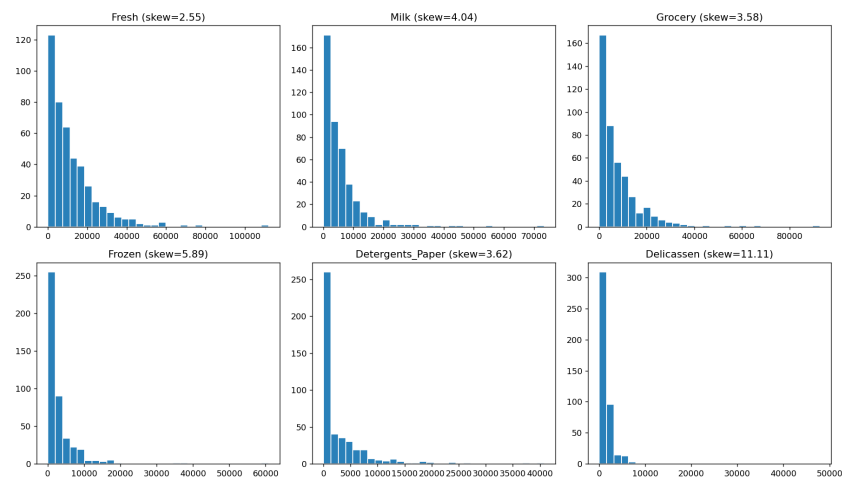


Figure 1. Raw spending distributions -- all six categories show severe right-skew.

2.2 Feature Strategy

Channel and Region were excluded from the clustering inputs and retained only for post-hoc validation, so the clustering discovers spending-pattern structure rather than trivially reproducing the existing 2-category Channel split.

2.3 Transformation and Scaling

log1p was applied to compress the skew (safely handling near-zero values), followed by StandardScaler to equalize scale across the six dimensions -- both necessary since K-Means is sensitive to scale and outliers.

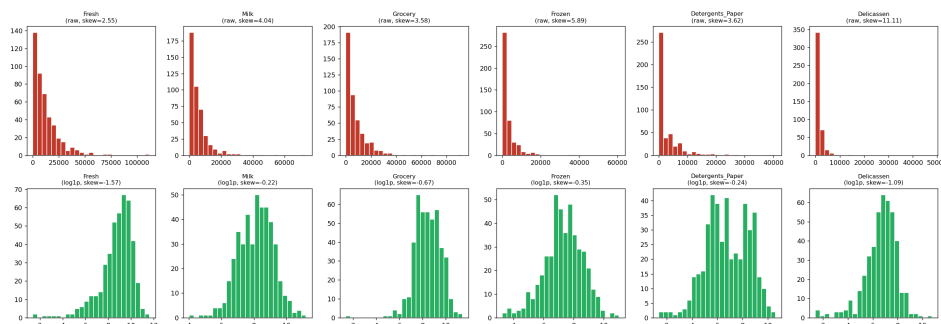


Figure 2. Before/after log1p transformation -- skewness dropped from 2.5-11.1 to near-symmetric (-1.6 to -0.2).

2.4 Cluster Count Selection

The elbow method curve was smooth without one clear bend, so silhouette score (a more objective measure) drove the decision. Silhouette peaked at $k=2$ (0.290), but $k=3$ was a close second (0.259) and scores dropped sharply for $k \geq 4$. $k=3$ was selected: it avoids trivially reproducing the binary Channel split while remaining statistically well-supported and providing richer, more actionable business segments.

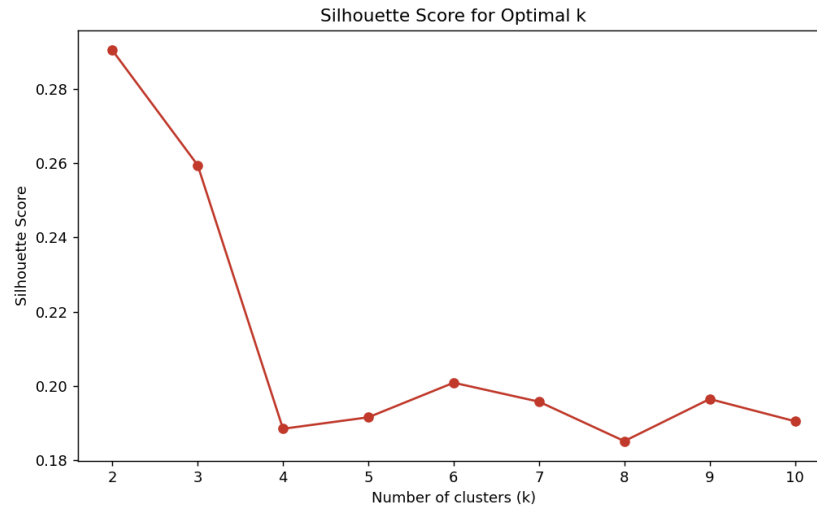


Figure 3. Silhouette score by k -- $k=3$ selected as the best balance of separation and segment richness.

3. Findings

3.1 Algorithm Comparison and Stability

K-Means and Agglomerative Clustering (Ward linkage) were both fit at $k=3$. Silhouette scores were nearly identical (K-Means 0.2594, Agglomerative 0.2547), but the cross-tabulation showed only partial agreement: one large group of about 200 customers was consistently identified by both methods, while the remaining customers split differently depending on the algorithm. This is reported honestly as a genuine stability limitation rather than an overstated claim of crisp, algorithm-agnostic segments. K-Means was used as the primary clustering going forward.

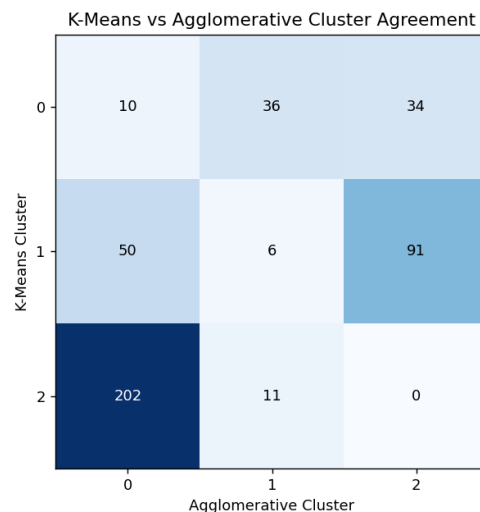


Figure 4. K-Means vs Agglomerative agreement -- one large segment is consistently found by both methods.

3.2 Cluster Visualization (PCA)

The first two principal components explain 71.3% of the variance in the transformed spending data (PC1: 44.1%, PC2: 27.2%), providing a reasonably faithful 2D view of the underlying 6-dimensional structure.

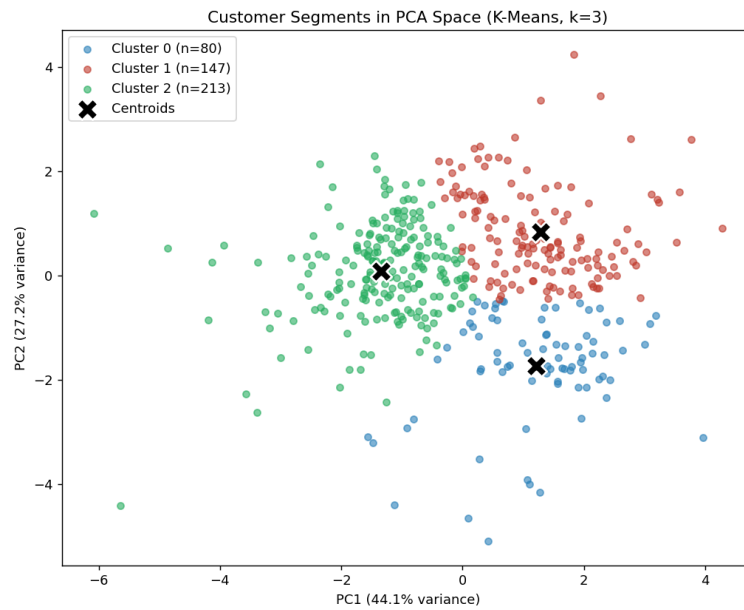


Figure 5. Customer segments in PCA space, with K-Means centroids marked.

3.3 Cluster Profiles

Cluster	Size	Fresh	Milk	Grocery	Frozen	Det./Paper	Delicassen
0	80	2,899	7,136	12,570	607	5,554	783
1	147	17,043	10,560	13,334	4,133	4,987	2,847
2	213	11,939	2,006	2,502	3,266	424	891



Figure 6. Normalized cluster spending profiles (original-unit averages shown in the table above).

3.4 Validation Against Channel and Region

Cluster 2 aligns very strongly with the Horeca channel (210 of 213 customers, ~99%) -- strong evidence this segment reflects genuine business structure rather than algorithmic noise. Cluster 0 leans Retail (48/80, 60%) and Cluster 1 is mixed but retail-leaning (91/147, 62%). Region showed no strong relationship with any cluster, suggesting spending behavior is driven by business type/need rather than geography.

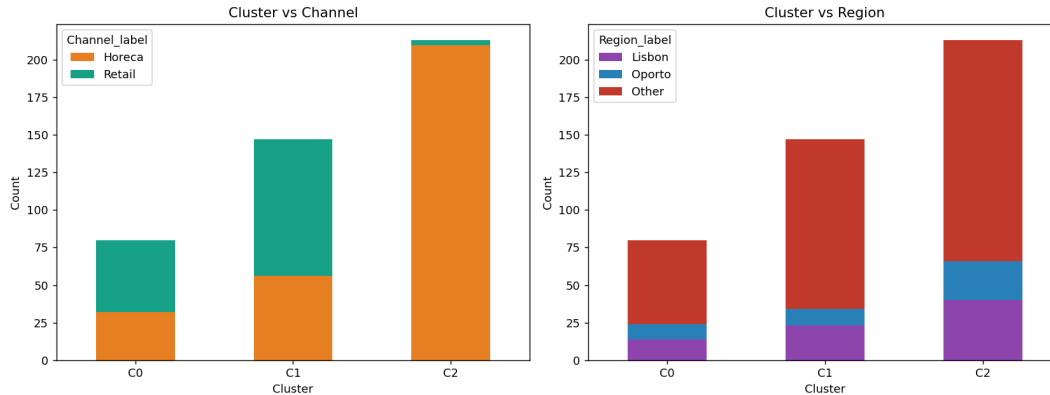


Figure 7. Cluster composition by Channel (left) and Region (right).

3.5 Business Segments and Recommended Actions

Cluster	Label	Recommended Action
0 (n=80)	Grocery & Household Retailers	Bulk grocery/detergent discounts and loyalty pricing on household staples.
1 (n=147)	High-Volume Bulk Buyers	Priority account management and volume-based contracts given broad, large spend.
2 (n=213)	Fresh-Focused Food Service (Horeca)	Fast, frequent fresh-product delivery scheduling; restaurant/hotel-specific promotions.

4. Limitations

- 1 K-Means and Agglomerative Clustering showed only partial agreement; one large segment is consistently found by both methods, but the boundary between the remaining two is less crisp.
- 2 With only 440 customers, smaller or rarer purchasing patterns may not form their own distinct, statistically detectable cluster.
- 3 The PCA visualization captures 71.3% of variance in 2D -- a useful but imperfect simplification of the true 6-dimensional structure.

5. Conclusion

Six skewed annual-spending categories were log-transformed, scaled, and clustered into three segments using K-Means, cross-validated against Agglomerative Clustering for stability. The Fresh-Focused Food Service segment shows a very strong, meaningful alignment with the existing Horeca channel (99%), giving confidence the clusters reflect real business structure, though the boundaries between the other two segments are less crisp -- a limitation reported honestly rather than overstated. Each segment has a distinct spending profile and a

corresponding recommended action, and a reusable assignment function is included to classify new customers using the saved preprocessing and clustering objects.

Dataset citation: Cardoso, M. (2014). Wholesale customers [Dataset]. UCI Machine Learning Repository.
<https://doi.org/10.24432/C5030X>